

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and deployed **Python**-based machine learning models for anomaly detection on large-scale time-series telemetry data, achieving an **~0.92 AUC** on held-out test sets.
- Developed and optimized **Python** data pipelines for ETL, processing approximately **3,000 events per second** to support AI and machine learning initiatives.
- Engineered **Java Spring Boot** REST APIs for integrating AI capabilities into existing operational systems, managing request validation and asynchronous processing.
- Managed production environments for enterprise applications, focusing on reliability and performance tuning using **Splunk** and **Dynatrace**, resolving critical incidents within SLA targets.
- Collaborated with operations teams to translate complex operational challenges into data-driven solutions, defining requirements for new monitoring and automation features.
- Implemented and maintained **Docker** containers for CI/CD pipelines using **Jenkins**, ensuring consistent deployment of applications and ML models in staging and production.

SKILLS

AI/ML – Machine Learning, Anomaly Detection, Predictive Modeling, Optimization, Scikit-Learn, PyTorch, TensorFlow, Data Modeling, Time-Series Analysis

Programming Languages – Python, JavaScript, TypeScript, Java, R

Backend Development – Spring Boot, Node.js, REST APIs, Flask, FastAPI

Data & Databases – SQL, PostgreSQL, MySQL, Oracle, Data Warehousing, Power Query, DAX, Tableau

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Kubernetes, Jenkins, Git, Linux

Observability & Operations – Splunk, Dynatrace, CI/CD, Incident Response, Change Management, GDPR

PROJECTS

Production Anomaly Detection System (HCL Technologies)

- Developed a **Python**-based anomaly detection system using **scikit-learn** to monitor operational telemetry data from enterprise applications, identifying over **500 unusual patterns** in its first month.
- Engineered feature pipelines that extracted relevant metrics from time-series data, feeding into the trained classification model for real-time analysis.
- Integrated model inference endpoints as **REST APIs**, enabling downstream systems to query for anomalies and trigger automated alerts, reducing manual review effort by approximately **30%**.

Data Pipeline Optimization for Operational Dashboards (HCL Technologies)

- Refactored and optimized existing **Python** ETL processes for a financial-services operations dashboard, improving data ingestion throughput by **22%**.
- Designed and implemented a new data schema in **PostgreSQL** to better support complex analytical queries and reporting requirements from business stakeholders.
- Created automated data validation checks within the pipeline to ensure data integrity, reducing data discrepancy issues reported by the operations team by roughly **15%**.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)